Physics-Informed Deep Learning for Gravitational Lensing Inverse Problems

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Reconstructing the convergence field of gravitational lenses from observed images is a severely ill-posed inverse problem. Standard deep learning approaches treat this as purely data-driven regression, ignoring the physics of light deflection. We introduce a physics-informed U-Net that embeds the thin-lens equation into the training loss via a differentiable spectral Poisson solver. On a synthetic dataset of SIS+NFW lensing systems, the physics-informed model achieves a convergence MSE of 1.75×10^{-4}, about 3.3 times lower than the baseline's 5.75×10^{-4}, and a re-lensing MSE of 8.0×10^{-6}, about 5.3 times lower than the baseline's 4.2×10^{-5}. The physics-informed reconstructions are also qualitatively cleaner, with fewer artifacts and better morphological fidelity. Under additive Gaussian noise the physics-informed model degrades faster than the baseline, with the two crossing at $\sigma \approx 0.04$. We attribute this to the model's reliance on fine image structure that noise destroys. These results show that embedding lensing physics into the training loop improves both accuracy and physical consistency of mass map reconstructions.

I. INTRODUCTION

Gravitational lensing, the deflection of light by intervening mass, is a powerful probe for mapping dark matter and constraining cosmological parameters [3, 4]. Strong lensing systems produce multiple images, arcs, or Einstein rings of background sources, encoding detailed information about the lens's projected mass distribution. Extracting this information requires solving an inverse problem: recovering the convergence field $\kappa(\boldsymbol{\theta})$ from the distorted image.

This inverse problem is ill-posed. Distinct mass distributions can produce nearly identical lensed images once noise and finite resolution are accounted for, and fundamental degeneracies such as the mass-sheet degeneracy [17, 18] prevent unique reconstruction without additional data. Traditional parametric approaches [6, 7, 24] impose strong assumptions on the mass profile and scale poorly to the large samples expected from LSST [19] and Euclid [20].

Deep learning offers an alternative: train a CNN to map directly from observed images to convergence fields, bypassing iterative fitting at inference time [8–10, 12]. However, most existing approaches treat this as generic image-to-image regression with pixel-wise loss functions, providing the network no information about the physics relating mass to image. The resulting predictions can contain features inconsistent with the lens equation.

We address this by embedding the thin-lens equation directly into the loss function. Given a predicted $\hat{\kappa}$, we solve the Poisson equation for the lensing potential using a differentiable spectral solver, compute the deflection field, and re-lens a source through the predicted mass map. The mismatch between the re-lensed and observed images provides an additional training signal that encourages physically consistent reconstructions.

Our contributions are: (i) a fully differentiable forward lensing pipeline that integrates into standard training loops; (ii) a composite loss balancing pixel-wise reconstruction with physics-based lensing consistency; (iii) demonstration that this approach yields lower error on both convergence and re-lensing metrics compared to a baseline U-Net; and (iv) analysis of failure modes, including a noise sensitivity tradeoff.

II. BACKGROUND

A. Gravitational lensing theory

In the thin-lens approximation, the lens equation relates image-plane coordinates $\boldsymbol{\theta}$ to source-plane coordinates $\boldsymbol{\beta}$ via

$$\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta}), \tag{1}$$

where the deflection angle $\boldsymbol{\alpha} = \nabla_{\boldsymbol{\theta}}\psi$ is the gradient of the lensing potential ψ. The potential satisfies the 2D Poisson equation

$$\nabla^2\psi(\boldsymbol{\theta}) = 2\kappa(\boldsymbol{\theta}), \tag{2}$$

with convergence $\kappa = \Sigma/\Sigma_{\rm cr}$, where Σ is the projected surface mass density and $\Sigma_{\rm cr} = (c^2/4\pi G)(D_s/D_l D_{ls})$ is the critical density.

The forward problem is straightforward: given κ and a source $I_s(\boldsymbol{\beta})$, solve for $\boldsymbol{\alpha}$ and evaluate $I_{\rm obs}(\boldsymbol{\theta}) = I_s(\boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta}))$. The inverse problem of recovering κ from $I_{\rm obs}$ is ill-posed because the source is unknown, the lens mapping is non-injective, and noise degrades the data.

B. Related work

U-Net architectures [13] have been applied to lensing reconstruction with encouraging results [10, 12]. Physics-informed neural networks (PINNs) [14, 15] embed PDEs into loss functions but have seen limited use in astrophysical inverse problems. Our approach differs from standard PINNs in that we do not enforce a PDE pointwise. Instead, we use the Poisson equation as part of

a differentiable forward model and penalize mismatch in observable space.

III. METHODOLOGY

A. Data generation

Each lens is modeled as an SIS with a simplified NFW subhalo. The SIS convergence is

$$\kappa_{\mathrm{SIS}}(\boldsymbol{\theta}) = \frac{\theta_E}{2|\boldsymbol{\theta} - \boldsymbol{\theta}_c|}, \tag{3}$$

with Einstein radius θ_E and center $\boldsymbol{\theta}_c$. The NFW component is computed from a projected profile with concentration and scale radius drawn from astrophysically motivated ranges. Source images consist of elliptical Gaussian blobs. Lensed images are generated on a 128×128 grid using bilinear interpolation. The dataset has 1000 samples (800 train, 200 validation), each a pair $(I_{\mathrm{obs}}, \kappa_{\mathrm{true}})$ normalized to $[0, 1]$. No noise or PSF is applied at generation time.

B. Spectral Poisson solver

In Fourier space the Poisson equation becomes algebraic:

$$\hat{\psi}(\mathbf{k}) = -\frac{2\hat{\kappa}(\mathbf{k})}{|\mathbf{k}|^2}, \quad \mathbf{k} \neq \mathbf{0}, \tag{4}$$

with $\hat{\psi}(\mathbf{0}) = 0$. The deflection is obtained via $\hat{\alpha}_j(\mathbf{k}) = ik_j\hat{\psi}(\mathbf{k})$. All operations (FFT, division, IFFT, gradient) are implemented in PyTorch and are fully differentiable. The spectral method assumes periodic boundary conditions, which introduces small edge artifacts for our grids since κ decays to near zero at the boundaries.

C. Architecture

Both models use a U-Net with four encoder blocks ($1 \to 32 \to 64 \to 128 \to 256$ channels, each with two 3×3 convolutions + ReLU + 2×2 max pool) and a symmetric decoder with skip connections and transpose convolutions. The output layer is a 1×1 convolution with sigmoid activation. The only difference between the baseline and physics-informed models is the loss function.

D. Loss function

The baseline uses pixel-wise MSE:

$$\mathcal{L}_{\mathrm{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(\hat{\kappa}_i - \kappa_i^{\mathrm{true}})^2. \tag{5}$$

Algorithm 1 Physics-informed training step

Require: Image I_{obs}, true κ_{true}, source I_s, weight λ

$\hat{\kappa} \leftarrow f_\theta(I_{\mathrm{obs}})$
$\mathcal{L}_{\mathrm{MSE}} \leftarrow \|\hat{\kappa} - \kappa_{\mathrm{true}}\|^2$
$\hat{\psi} \leftarrow \texttt{FFT_Poisson}(\hat{\kappa})$
$\hat{\boldsymbol{\alpha}} \leftarrow \nabla\hat{\psi}$
$\hat{I}_{\text{re-lensed}} \leftarrow \texttt{grid_sample}(I_s,\ \boldsymbol{\theta} - \hat{\boldsymbol{\alpha}})$
$\mathcal{L}_{\mathrm{lens}} \leftarrow \|\hat{I}_{\text{re-lensed}} - I_{\mathrm{obs}}\|^2$
$\mathcal{L} \leftarrow \mathcal{L}_{\mathrm{MSE}} + \lambda\,\mathcal{L}_{\mathrm{lens}}$
Backpropagate $\mathcal{L}$; update θ

The physics-informed model adds a lensing consistency term. Given $\hat{\kappa}$, we solve for $\hat{\psi}$, compute $\hat{\boldsymbol{\alpha}} = \nabla\hat{\psi}$, apply the lens equation $\hat{\boldsymbol{\beta}} = \boldsymbol{\theta} - \hat{\boldsymbol{\alpha}}$, and re-lens the source:

$$\mathcal{L}_{\mathrm{lens}} = \frac{1}{N}\sum_{i=1}^{N}(\hat{I}_{\text{re-lensed},i} - I_{\mathrm{obs},i})^2. \tag{6}$$

The total loss is

$$\mathcal{L} = \mathcal{L}_{\mathrm{MSE}} + \lambda\,\mathcal{L}_{\mathrm{lens}}, \tag{7}$$

with $\lambda = 0.1$. For this study we use the known source to isolate the effect of the physics loss from source reconstruction errors.

E. Training

Both models are trained for 40 epochs with Adam [16] (learning rate 10^{-3}, batch size 8) on a single NVIDIA GPU. One epoch takes about 15 s for the baseline and 25 s for the physics-informed model.

IV. RESULTS

A. Training dynamics

Figure 1 shows the baseline loss curves. Both train and validation loss drop rapidly in the first few epochs and plateau near 3–4×10^{-4}, with mild validation spikes around epochs 22–26.

The physics-informed model (Fig. 2, MSE component only) starts higher because the lensing loss competes with the pixel-wise term. The training MSE decreases steadily over all 40 epochs, ultimately falling below the baseline's final value. The validation MSE shows more oscillation in the first 20 epochs but converges to a similar level.

B. Baseline reconstruction quality

Figure 3 shows the baseline model's predictions on a validation sample at four training stages. Each row of panels shows, left to right, the lensed image, true κ, and predicted κ.

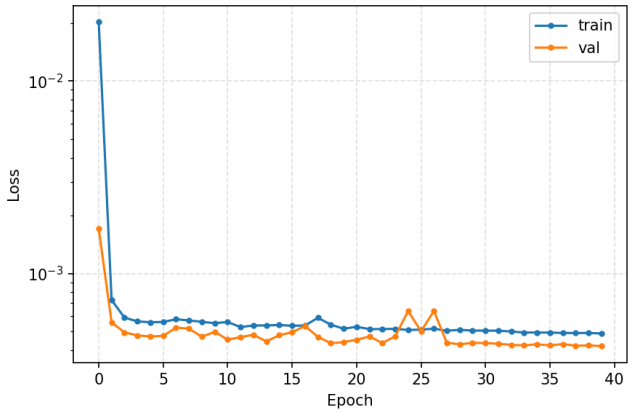

FIG. 1. Baseline U-Net loss curves. Train and validation loss plateau near 3–4 $\times 10^{-4}$, with mild validation spikes.

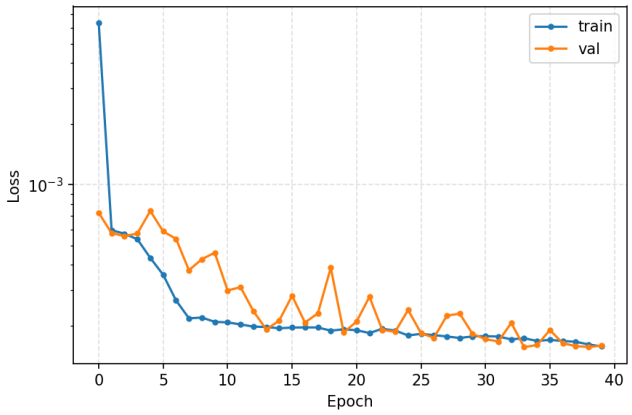

FIG. 2. Physics-informed U-Net loss curves (MSE component). The training loss decreases steadily and ultimately falls below the baseline's plateau.

At epoch 10, the predicted κ has a strong teal background offset and the mass concentrations appear as diffuse blobs with checkerboard texture from the transpose convolutions. By epoch 20, the background shifts but blocky rectangular artifacts dominate. At epoch 30, the mass structures are better defined but the blocky patchwork persists. By epoch 40, the artifacts have smoothed somewhat but the reconstruction remains noisy with residual background offset.

C. Physics-informed reconstruction quality

Figure 4 shows the physics-informed model at five training stages. Each row of panels shows the lensed image, true κ, predicted κ, and re-lensed image.

At epoch 1, the predicted κ has a strong teal background with a grid-like pattern and blob-shaped mass concentrations. Despite this, the re-lensed image already shows rough structural agreement with the observation.

TABLE I. Validation set metrics (mean over 200 samples).

Metric	Baseline	Physics-Informed
κ MSE	5.75×10^{-4}	1.75×10^{-4}
Re-lensing MSE	4.2×10^{-5}	8.0×10^{-6}

By epoch 10, the background is still elevated and ring artifacts appear around the mass peaks, a signature of the spectral solver's periodic boundary conditions interacting with bilinear sampling. The re-lensed image matches the input well.

At epoch 20, the background has darkened substantially and the rings have shrunk. By epoch 30, only faint ring traces remain and the mass peak is well-localized. At epoch 40, the reconstruction is clean: the mass concentration has good amplitude and position, the background is dark, and the re-lensed image closely reproduces the observed morphology.

D. Side-by-side comparison

Figure 5 compares both models on a validation sample. The panels show, left to right: input lensed image, true κ, baseline κ, physics-informed κ, input lensed image (repeated), baseline re-lensed image, physics-informed re-lensed image.

The baseline κ is dominated by blocky checkerboard artifacts with a large background offset. The physics-informed κ is much cleaner, with a compact, well-localized mass peak and only faint ring artifacts. Both re-lensed images reproduce the input morphology, but the physics-informed version is more faithful in position and flux.

E. Quantitative metrics

Table I reports validation-set averages. The physics-informed model outperforms the baseline on both metrics: 3.3× lower convergence MSE and 5.3× lower re-lensing MSE. The improvement on convergence MSE shows that the physics constraint acts as an effective regularizer beyond just optimizing for re-lensing consistency.

F. Noise robustness

We add Gaussian noise with $\sigma \in \{0.0, 0.02, 0.05\}$ to input images at test time (models trained on clean data). Figure 6 shows the results.

At $\sigma = 0$ the physics-informed model has the lower error, consistent with Table I. However, it degrades more steeply: its MSE rises from $\sim 1 \times 10^{-4}$ to $\sim 7 \times 10^{-3}$ at $\sigma = 0.05$, while the baseline rises from $\sim 5 \times 10^{-4}$ to $\sim 2.5 \times 10^{-3}$. The curves cross at $\sigma \approx 0.04$.

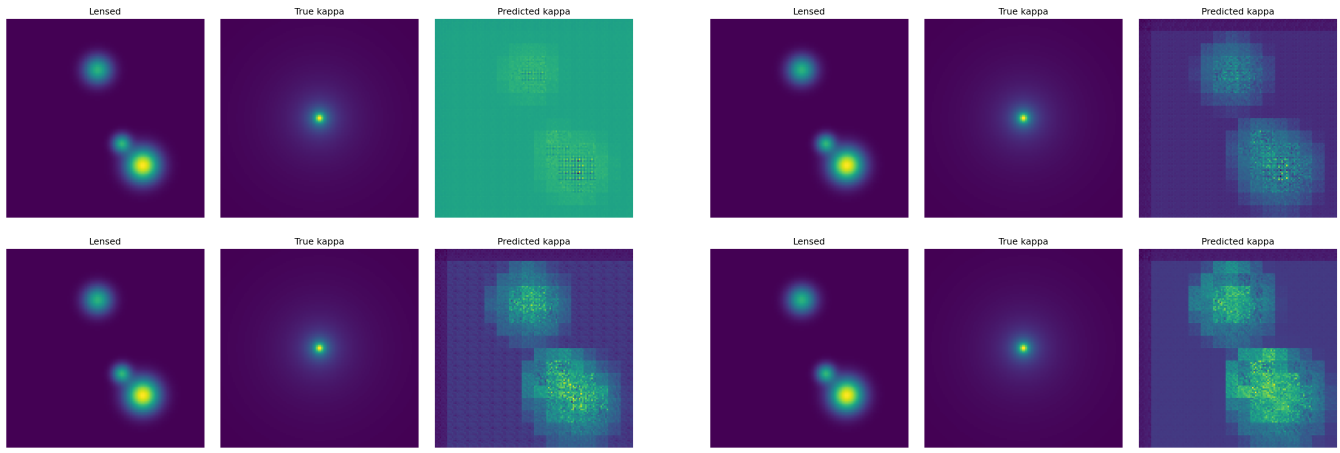

FIG. 3. Baseline predictions at epochs 10 (top left), 20 (top right), 30 (bottom left), and 40 (bottom right). Each panel: lensed image, true κ, predicted κ. Checkerboard and blocky artifacts persist throughout, with a residual background offset.

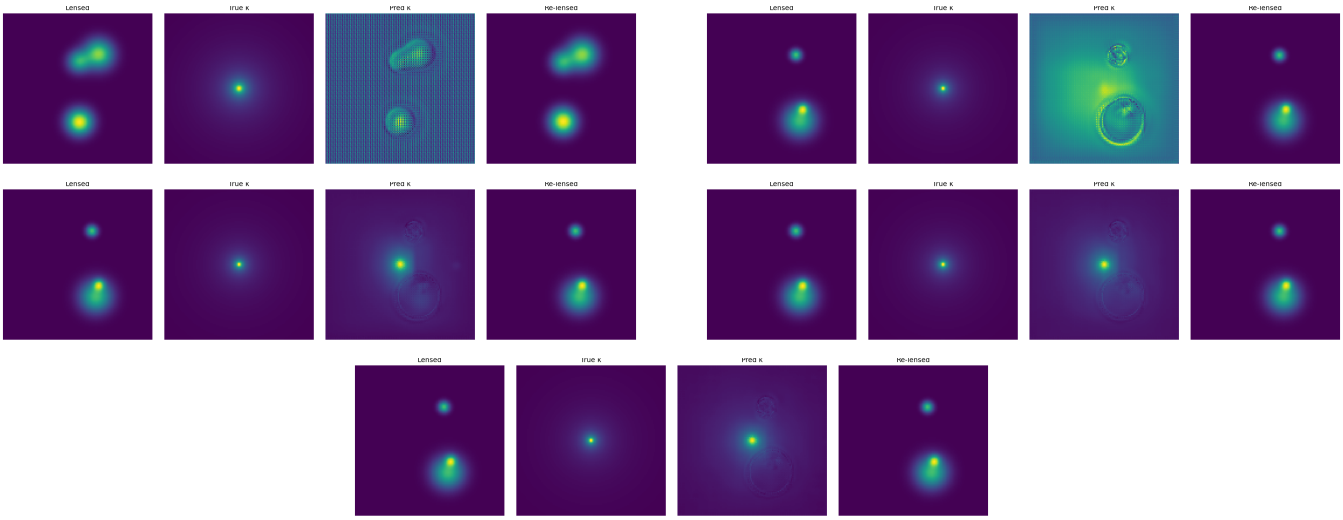

FIG. 4. Physics-informed predictions at epochs 1 (top left), 10 (top right), 20 (middle left), 30 (middle right), and 40 (bottom). Each panel: lensed image, true κ, predicted κ, re-lensed image. Ring artifacts from the spectral solver fade over training. By epoch 40 the reconstruction is clean and the re-lensed image matches the observation well.

This happens because the physics-informed model relies on fine image structure to constrain the mass map through the forward lensing pipeline. Noise corrupts these features and degrades the physics loss gradients. The baseline learns a more conservative mapping from coarser features and is less sensitive to high-frequency perturbations.

V. DISCUSSION

A. Why the physics loss helps

The physics loss constrains the network output to the manifold of convergence fields consistent with the observed image under the lens equation. This acts as a physics-based regularizer that reduces the effective solution space. The analogy is to compressed sensing, where structural priors enable reconstruction from fewer measurements. Here the prior is lensing consistency rather than sparsity.

The regularization effect is visible in both metrics. The re-lensing improvement is expected by construction, but the convergence MSE improvement shows the physics loss guides the network toward solutions that are better overall, not merely shifted between objectives.

B. Artifacts

The baseline suffers from checkerboard artifacts from the transpose convolutions and a persistent background offset. These are standard encoder-decoder issues, made worse here by the lack of any structural constraint beyond

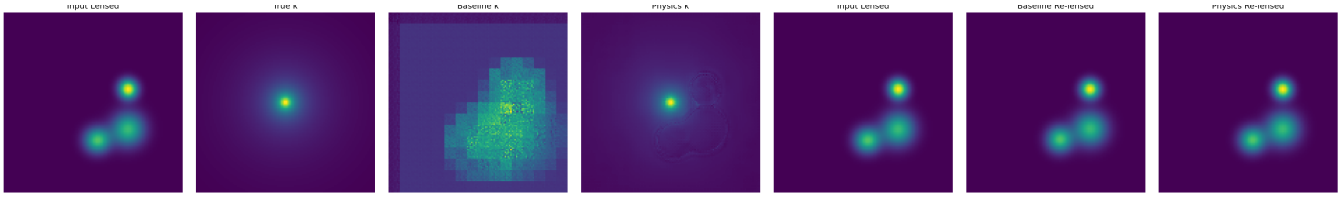

FIG. 5. Side-by-side comparison on a validation sample. Left to right: input lensed image, true κ, baseline κ, physics-informed κ, input lensed image, baseline re-lensed, physics-informed re-lensed. The baseline κ shows severe checkerboard artifacts; the physics-informed κ is clean with a well-localized mass peak.

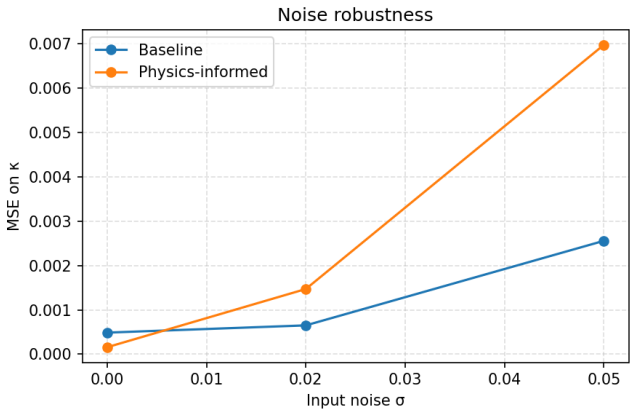

FIG. 6. Convergence MSE vs. input noise σ. The physics-informed model (orange) starts lower but degrades faster, crossing the baseline (blue) near $\sigma \approx 0.04$.

pixel-wise matching.

The physics-informed model shows ring-like artifacts from the spectral solver's periodic boundary conditions interacting with bilinear grid sampling. These are strongest in early epochs and fade as training progresses, suggesting the network learns to produce κ fields that minimize them. Possible mitigations include low-pass filtering before the Poisson solve, finite-difference solvers, or zero-padding.

C. Limitations

The lens models (SIS + NFW) are idealized; real lenses have irregular mass distributions with baryonic substructure and line-of-sight contamination. The source is assumed known; joint source-lens reconstruction would require a more sophisticated setup, possibly with an adversarial or variational component. The dataset is small (1000 samples), and the physics loss weight λ was not systematically tuned. The noise sensitivity of the physics-informed model needs to be addressed before application to real data, for instance through noise-augmented training or a noise-aware physics loss.

VI. CONCLUSION

We have presented a physics-informed U-Net for gravitational lens mass reconstruction that embeds the thin-lens equation into the training loss via a differentiable spectral Poisson solver. On synthetic SIS+NFW data, the physics-informed model achieves 3.3× lower convergence MSE and 5.3× lower re-lensing MSE than the baseline, with qualitatively cleaner reconstructions. The physics loss acts as a regularizer that improves both pixel-level accuracy and physical consistency.

The main limitation is reduced noise robustness: the physics-informed model degrades faster under input noise, crossing below the baseline at $\sigma \approx 0.04$. Addressing this through noise augmentation is a natural next step. Other extensions include more realistic lens models, joint source-lens reconstruction, and application to real data from HST, JWST, or LSST.

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